

ARRAYS

1. Find the Second Largest Element

Description

Given an array of integers, find the second largest distinct element in the array.

Input Format

First line contains an integer N.

Second line contains N space-separated integers.

Output Format

Print the second largest distinct element.

Sample Input

```
6
12 35 1 10 34 35
```

Sample Output

```
34
```

Explanation

The largest element is 35 and the second largest distinct element is 34.

OUTPUT :

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n,arr[50],max1=0,max2=0;
```

```
    printf("Enter Number :");
```

```
    scanf("%d",&n);
```

```
    for(int i=0;i<n;i++){
```

```
        scanf("%d",&arr[i]);
```

```
    }
```

```
    for(int i=1;i<n;i++){
```

```
        if(max1<arr[i]){
```

```
            max1=arr[i];
```

```
            max2=max1;
```

```
        }
```

```
        if(max2<arr[i] && max1>arr[i]){
```

```
            max2=arr[i];
```

```
        }
```

```
    }
```

```
    printf("%d",max2);
```

```
    return 0;
```

```
}
```

2. Move All Zeros to the End

Description

Move all zero elements to the end while maintaining the order of non-zero elements.

Input Format

N

N space-separated integers

Output Format

Print the modified array.

Sample Input

7

0 1 0 3 12 0 5

Sample Output

1 3 12 5 0 0 0

Explanation

All non-zero elements retain their order, and zeros are moved to the end.

OUTPUT :

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n,a[100],c=0;
```

```
    printf("Enter Number :");
```

```
    scanf("%d",&n);
```

```
    for(int i=0;i<n;i++){  
        scanf("%d",&a[i]);
```

```
    }
```

```
    for(int i=0;i<n;i++){
```

```
        if(a[i]!=0){
```

```
            printf("%d",a[i]);
```

```
            c++;
```

```
        }
```

```
    }
```

```
    while(c<n){
```

```
        printf("0");
```

```
        c++;
```

```
    }
```

```
    return 0;
```

```
}
```

3. Check if Array is Sorted

Description

Determine whether the array is sorted in ascending order.

Input Format

N

N space-separated integers

Output Format

Print YES if sorted, otherwise NO.

Sample Input

5

2 4 6 8 10

Sample Output

YES

Explanation

Every element is greater than or equal to the previous element.

OUTPUT :

```

#include <stdio.h>

int main()
{
    int n,a[100],fl=0;
    printf("Ennter Number :");
    scanf("%d",&n);

    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    int s=a[0];
    for(int i=0;i<n;i++){
        if(a[i]>=s){
            s=a[i];
            fl=1;
        }
        else{
            fl=0;
            break;
        }
    }
    if(fl==1){
        printf("yes\n");
    }
    else{
        printf("no\n");
    }
    return 0;
}

```

4. Find Missing Number

Description

An array contains numbers from 1 to N with one number missing. Find the missing number.

Input Format

N

N-1 space-separated integers

Sample Input

5

1 2 3 5

Sample Output

4

Explanation

Numbers from 1 to 5 should be present. Number 4 is missing.

OUTPUT :

```

#include <stdio.h>

int main()
{
    int n,sum=0,mis;
    printf("ENTER NUMBER :");
    scanf("%d",&n);

```

```

int arr[n-1];
for(int i=0;i<n-1;i++){
    scanf("%d",&arr[i]);
    sum=sum+arr[i];
}
mis=(n*(n+1))/2-sum;
printf("%d",sum);
return 0;
}

```

5. Rotate Array by K Positions

Description

Rotate the array to the right by K positions.

Input Format

N

N space-separated integers

K

Sample Input

5

1 2 3 4 5

2

Sample Output

4 5 1 2 3

Explanation

After rotating twice to the right, the last two elements move to the front.

OUTPUT :

```
#include <stdio.h>
```

```

int main() {
    int n;
    if(scanf("%d",&n)!=1);

```

```

    int arr[n];
    for(int i =0;i<n;i++){
        scanf("%d",&arr[i]);
    }

```

```

    int k;
    scanf("%d",&k);
    int temp[n];
    for(int i=0;i<n;i++){
        temp[(i+k)%n]=arr[i];
    }
    for(int i=0;i<n;i++) {
        printf("%d",temp[i]);
        if(i<n-1){
            printf(" ");
        }
    }
    printf("\n");

```

```
return 0;
```

```
}
```

6. Remove Duplicates from a Sorted Array

Description

Remove duplicate elements from a sorted array.

Input Format

N

N space-separated integers

Sample Input

8

1 1 2 2 3 4 4 5

Sample Output

1 2 3 4 5

Explanation

Duplicate values are removed, leaving only unique elements.

OUTPUT :

```
#include <stdio.h>
```

```
int main() {
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;i++) {
        scanf("%d",&arr[i]);
    }
    if (n==0){
        return 0;
    }
    printf("%d", arr[0]);
    for(int i=1;i<n;i++) {
        if(arr[i]!=arr[i-1]){
            printf(" %d",arr[i]);
        }
    }
    printf("\n");

    return 0;
}
```

7. Find Leaders in an Array

Description

An element is a leader if it is greater than all elements to its right.

Input Format

N

N space-separated integers

Sample Input

6

16 17 4 3 5 2

Sample Output

17 5 2

Explanation

17 is greater than all elements to its right.

5 is greater than 2.

2 is always a leader because it is the last element.

OUTPUT :

```
#include <stdio.h>
```

```
int main() {
    int t;
    scanf("%d",&t);

    while (t--) {
        int x,y;
        scanf("%d%d",&x,&y);

        if(x>=y){
            printf("0\n");
        } else {
            int diff=y-x;
            int games=diff/8;
            if(diff%8!=0){
                games++;
            }
            printf("%d\n",games);
        }
    }

    return 0;
}
```

8. Find Frequency of Each Element

Description

Print the frequency of each distinct element.

Input Format

N

N space-separated integers

Sample Input

7

1 2 2 3 1 4 2

Sample Output

1 -> 2

2 -> 3

3 -> 1

4 -> 1

Explanation

Count the occurrences of each element.

OUTPUT :

```
#include <stdio.h>
```

```
int main(){
    int n;
    scanf("%d",&n);

    int arr[n];
    for(int i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
}
```

```

    }

    int visit[n];
    for(int i=0;i<n;i++){
        visit[i] =0;
    }

    for(int i=0;i<n;i++){
        if(visit[i]==1){
            continue;
        }

        int count =1;
        for(int j=i+1;j<n;j++){
            if(arr[i]==arr[j]){
                count++;
                visit[j] =1;
            }
        }

        printf("%d %d\n",arr[i],count);
    }

    return 0;
}

```

9. Find Pair with Given Sum

Description

Check whether any pair of elements adds up to a given target.

Input Format

N

N space-separated integers

Target

Sample Input

5

2 7 11 15 3

9

Sample Output

YES

Explanation

2 + 7 = 9.

OUTPUT :

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n,a[50],target,sum=0,i;
```

```
    scanf("%d",&n);
```

```
    for(int i=0;i<n;i++){
```

```
        scanf("%d",&a[i]);
```

```
    }
```

```
    scanf("%d",&target);
```

```
    for(int i=0;i<n;i++){
```

```

        for(int j=i+1;j<n;j++){

if(a[i]+a[j]==target){
    printf("YES\n");
    return 0;
}

}

}

printf("NO\n");
return 0;
}

```

10. Maximum Difference

Description

Find the maximum difference between two elements where the larger element comes after the smaller one.

Sample Input

7

2 3 10 6 4 8 1

Sample Output

8

Explanation

Maximum difference is $10 - 2 = 8$.

OUTPUT:

```
#include <stdio.h>
```

```

int main()
{
    int n,a[50],i,j,max,s,min;
    scanf("%d",&n);

    for(i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    max=a[0];
    for(i=1;i<n;i++){
        if(a[i]>max){
            max=a[i];
            s=i;
        }
    }
    min=a[0];
    for(int i=0;i<s;i++){
        if(a[i]<min){
            min=a[i];
        }
    }
    printf("DIFF :%d ",max-min);
    // printf("%d ",min);7 2 3 10 6 4 8 1

    return 0;
}

```

11. Product of Array Except Self

Description

For every index, find the product of all elements except itself.

Sample Input

4

1 2 3 4

Sample Output

24 12 8 6

Explanation

For index 0: $2 \times 3 \times 4 = 24$

For index 1: $1 \times 3 \times 4 = 12$

and so on.

OUTPUT :

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n,a[50],i,j,p=1;
```

```
    scanf("%d",&n);
```

```
    for(i=0;i<n;i++){
```

```
        scanf("%d",&a[i]);
```

```
    }
```

```
    for(i=0;i<n;i++){
```

```
        p=p*a[i];
```

```
    }
```

```
    for(i=0;i<n;i++){
```

```
        printf("%d ",p/a[i]);
```

```
    }
```

```
    return 0;
```

```
}
```

13. Equilibrium Index

Description

Find an index where left sum equals right sum.

Sample Input

5

1 3 5 2 2

Sample Output

2

Explanation

Left sum = $1 + 3 = 4$

Right sum = $2 + 2 = 4$

OUTPUT :

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n,a[50],sum=0,lf=0,rf=0,ix=0;
```

```

scanf("%d",&n);

for(int i=0;i<n;i++){
    scanf("%d",&a[i]);
    sum=sum+a[i];
}
for(int i=0;i<n;i++){
    rf=sum-lf-a[i];
    if(lf==rf){
        ix=i;
        break;
    }
    lf+=a[i];
}
printf("%d",ix);

return 0;
}

```

14. Rearrange Positive and Negative Numbers

Description

Arrange positive and negative numbers alternately.

Sample Input

6

1 -2 3 -4 5 -6

Sample Output

1 -2 3 -4 5 -6

Explanation

The array is already in alternating order.

OUTPUT :

```
#include <stdio.h>
```

```

int main()
{
    int n,a[50],temp=0,target,c=0;
    scanf("%d",&n);

    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    for(int i=0;i<n;i++){
        for(int j=i+1;j<n;j++){
            if(a[i]>a[j]){
                temp=a[i];
                a[i]=a[j];
                a[j]=temp;
            }
        }
    }
    target=a[n/2];
    for(int i=0;i<n;i++){
        if(a[i]==target)
            c++;
    }
    if(c>n/2)

```

```

printf("\n%d",a[n/2]);
    else
printf("no major element");

    return 0;
}

```

15. Find the Union of Two Arrays

Description

Given two arrays, find all distinct elements present in either array.

Input Format

First line: N

Second line: N integers

Third line: M

Fourth line: M integers

Sample Input

```

5
1 2 3 4 5

```

```

4

```

```

1 2 6 7

```

Sample Output

```

1 2 3 4 5 6 7

```

Explanation

Union contains all unique elements from both arrays.

OUTPUT :

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n,a[50],m,a2[50],f;
    scanf("%d",&n);
```

```
    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
```

```
    scanf("%d",&m);
    for(int i=0;i<m;i++){
        scanf("%d",&a2[i]);
    }
```

```
    for(int i=0;i<n;i++){
        printf("%d",a[i]);
    }
```

```
    for(int j=0;j<m;j++){
        f=0;
        for(int i=0;i<n;i++){
            if(a[i]==a2[j]){
                f=1;
                break;
            }
        }
        if(f==0)
            printf("%d",a2[j]);
    }
```

```

    }
}

```

```
    return 0;
}
```

16. Find the Intersection of Two Arrays

Description

Find common elements present in both arrays.

Sample Input

5

1 2 3 4 5

5

3 4 5 6 7

Sample Output

3 4 5

Explanation

3, 4, and 5 are present in both arrays.

OUTPUT :

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n,a[50],m,a2[50],c;
    scanf("%d",&n);
```

```
    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
```

```
    scanf("%d",&m);
    for(int j=0;j<m;j++){
        scanf("%d",&a2[j]);
    }
```

```
    for(int i=0;i<n;i++){
        for(int j=0;j<m;j++){
            if(a[i]==a2[j]){
                printf("%d",a[i]);
                break;
            }
        }
    }
```

```
}
```

```
}
```

```
    return 0;
```

```
}
```

17. Find the First Repeating Element

Description

Find the first element that appears more than once.

Sample Input

7

10 5 3 4 3 5 6

Sample Output

5

Explanation

5 is the first element whose repetition occurs earliest.

OUTPUT:

```
#include <stdio.h>
#include <stdlib.h>
int main()
{
    int n,i,j;
    scanf("%d",&n);
    int arr=(int *)malloc(n(sizeof(int)));
    for(i=0;i<n;i++)
        scanf("%d",&arr[i]);
    for(i=0;i<n;i++){
        for(j=i+1;j<n;j++){
            if(arr[i]==arr[j]){
                printf("\n%d",arr[i]);
                return 0;
            }
        }
    }
    return 0;
}
```

18. Find the First Non-Repeating Element

Description

Find the first element that occurs exactly once.

Sample Input

6

9 4 9 6 7 4

Sample Output

6

Explanation

6 is the first element with frequency 1.

OUTPUT:

```
#include <stdio.h>
#include <stdlib.h>
int main()
{
    int n,i,j,f;
    scanf("%d",&n);
    int arr=(int *)malloc(n(sizeof(int)));
    for(i=0;i<n;i++)
        scanf("%d",&arr[i]);
    for(i=0;i<n;i++){
        f=0;
        for(j=0;j<n;j++){
            if(i!=j&&arr[i]==arr[j])
                f=1;
        }
        if(f==0){
            printf("\n%d",arr[i]);
            return 0;
        }
    }
}
```

```

}
}
return 0;
}

```

19. Maximum Consecutive Ones

Description

Find the maximum number of consecutive 1s in a binary array.

Sample Input

```

8
1 1 0 1 1 1 0 1

```

Sample Output

```

3

```

Explanation

The longest sequence of consecutive 1s has length 3.

OUTPUT:

```

#include <stdio.h>
#include <stdlib.h>
int main()
{
    int n,i,j,c=0,m=0;
    scanf("%d",&n);
    int arr=(int *)malloc(n*sizeof(int));
    for(i=0;i<n;i++)
        scanf("%d",&arr[i]);
    for(i=0;i<n;i++){
        if(arr[i]==1)
            c++;
        else{
            if(c>m)
                m=c;
            c=0;
        }
    }
    if(c>m)
        m=c;
    printf("\n%d",m);
}

```

20. Find All Duplicate Elements

Description

Print all elements that appear more than once.

Sample Input

```

8
1 2 3 1 2 4 5 2

```

Sample Output

```

1 2

```

Explanation

Elements 1 and 2 occur multiple times.

OUTPUT:

```

#include <stdio.h>

```

```
#include <stdlib.h>
int main()
{
    int n,i,j,temp,c;
    scanf("%d",&n);
    int arr=(int *)malloc(n(sizeof(int)));
    for(i=0;i<n;i++)
        scanf("%d",&arr[i]);
    for(i=0;i<n;i++){
        for(j=i+1;j<n;j++){
            if(arr[i]>arr[j]){
                temp=arr[i];
                arr[i]=arr[j];
                arr[j]=temp;
            }
        }
        for(i=0;i<n-1;i++){
            if(arr[i]==arr[i+1]){
                printf("%d ",arr[i]);
                while(i <n-1 && arr[i]==arr[i+1])
                    i++;
            }
        }
        return 0;
    }
```